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SUPPLEMENTAL METHODS

Search Strategy

PubMed:

#1	(“Glaucoma” OR “Glaucomatous” OR “Primary glaucoma” OR “Open-angle glaucoma” OR “Open angle glaucoma” OR “Close-angle glaucoma” OR “Close angle glaucoma” OR “Angle-closure glaucoma” OR “Angle closure glaucoma” OR “Narrow-angle glaucoma” OR “Narrow angle glaucoma” OR “Optic neuropathy” OR “Optic nerve disease” OR “Intra-ocular pressure” OR “Intra ocular pressure” OR “Cup-disc ratio” OR “Cup-to-disc ratio” OR “Cup to disc ratio” OR “Central Corneal Thickness” OR “Visual field” OR “Visual acuity”)
#2	(((“Chronic” OR “Long-term” OR “Long term” OR “Permanent” OR “Terminal” OR “Irreversible” OR “End-stage” OR “End stage”) AND (“Renal” OR “Kidney”) AND (“Disease” OR “Damage” OR “Dysfunction” OR “Deterioration” OR “Decline” OR “Insufficiency” OR “Impairment” OR “Malfunction” OR “Failure”)) OR “Nephropathy” OR “Glomerulopathy” OR “Albuminuria” OR "Albumin-creatinine ratio" OR "Albumin creatinine ratio" OR “Albumin-to-creatinine ratio” OR “Albumin to creatinine ratio” OR "Albumin-excretion rate" OR "Albumin excretion rate" OR “Proteinuria” OR “Protein-creatinine ratio" OR "Protein creatinine ratio" OR “Protein-to-creatinine ratio” OR “Protein to creatinine ratio” OR "Protein-excretion rate" OR "Protein excretion rate" OR “Estimated glomerular filtration rate” OR “Glomerular filtration rate” OR “Kidney transplant” OR “Renal transplant” OR “Kidney transplantation” OR “Renal transplantation” OR “Kidney replacement” OR “Renal replacement”)
#3	#1 AND #2

Embase:

#1	('glaucoma'/exp OR 'glaucoma' OR 'glaucomatous' OR 'primary glaucoma'/exp OR 'primary glaucoma' OR 'open-angle glaucoma'/exp OR 'open-angle glaucoma' OR 'open angle glaucoma'/exp OR 'open angle glaucoma' OR 'close-angle glaucoma' OR 'close angle glaucoma' OR 'angle-closure glaucoma'/exp OR 'angle-closure glaucoma' OR 'angle closure glaucoma'/exp OR 'angle closure glaucoma' OR 'narrow-angle glaucoma'/exp OR 'narrow-angle glaucoma' OR 'narrow angle glaucoma'/exp OR 'narrow angle glaucoma' OR 'optic neuropathy'/exp OR 'optic neuropathy' OR 'optic nerve disease'/exp OR 'optic nerve disease' OR 'intra-ocular pressure'/exp OR 'intra-ocular pressure' OR 'intra ocular pressure'/exp OR 'intra ocular pressure' OR 'cup-disc ratio'/exp OR 'cup-disc ratio' OR 'cup-to-disc ratio'/exp OR 'cup-to-disc ratio' OR 'cup to disc ratio'/exp OR 'cup to disc ratio' OR 'central corneal thickness'/exp OR 'central corneal thickness' OR 'visual field'/exp OR 'visual field' OR 'visual acuity'/exp OR 'visual acuity')
#2	(('chronic' OR 'long-term' OR 'long term' OR 'permanent' OR 'terminal' OR 'irreversible' OR 'end-stage'OR 'end

	stage') AND ('renal'/exp OR 'renal' OR 'kidney'/exp OR 'kidney') AND ('disease'/exp OR 'disease' OR 'damage' OR 'dysfunction' OR 'deterioration'/exp OR 'deterioration' OR 'decline'/exp OR 'decline' OR 'insufficiency' OR 'impairment'/exp OR 'impairment' OR 'malfunction' OR 'failure'/exp OR 'failure') OR 'nephropathy'/exp OR 'nephropathy' OR 'glomerulopathy'/exp OR 'glomerulopathy' OR 'albuminuria'/exp OR 'albuminuria' OR 'albumin-creatinine ratio'/exp OR 'albumin-creatinine ratio' OR 'albumin creatinine ratio'/exp OR 'albumin creatinine ratio' OR 'albumin-to-creatinine ratio'/exp OR 'albumin-to-creatinine ratio' OR 'albumin to creatinine ratio'/exp OR 'albumin to creatinine ratio' OR 'albumin-excretion rate' OR 'albumin excretion rate'/exp OR 'albumin excretion rate' OR 'proteinuria'/exp OR 'proteinuria' OR 'protein-creatinine ratio'/exp OR 'protein-creatinine ratio' OR 'protein creatinine ratio'/exp OR 'protein creatinine ratio' OR 'protein-to-creatinine ratio'/exp OR 'protein-to-creatinine ratio' OR 'protein to creatinine ratio'/exp OR 'protein to creatinine ratio' OR 'protein-excretion rate' OR 'protein excretion rate' OR 'estimated glomerular filtration rate'/exp OR 'estimated glomerular filtration rate' OR 'glomerular filtration rate'/exp OR 'glomerular filtration rate' OR 'kidney transplant'/exp OR 'kidney transplant' OR 'renal transplant'/exp OR 'renal transplant' OR 'kidney transplantation'/exp OR 'kidney transplantation' OR 'renal transplantation'/exp OR 'renal transplantation' OR 'kidney replacement' OR 'renal replacement')
#3	([article]/lim OR [article in press]/lim) AND [humans]/lim AND [english]/lim AND [embase]/lim NOT [medline]/lim
#4	#1 AND #2 AND #3

Cochrane Library:

#1	Glaucoma OR Glaucomatous OR Primary glaucoma OR Open-angle glaucoma OR Open angle glaucoma OR Close-angle glaucoma OR Close angle glaucoma OR Angle-closure glaucoma OR Angle closure glaucoma OR Narrow-angle glaucoma OR Narrow angle glaucoma OR Optic neuropathy OR Optic nerve disease OR Intra-ocular pressure OR Intra ocular pressure OR Cup-disc ratio OR Cup-to-disc ratio OR Cup to disc ratio OR Central Corneal Thickness OR Visual field OR Visual acuity
#2	Chronic OR Long-term OR Long term OR Permanent OR Terminal OR Irreversible OR End-stage OR End stage
#3	Renal OR Kidney
#4	Disease OR Damage OR Dysfunction OR Deterioration OR Decline OR Insufficiency OR Impairment OR Malfunction OR Failure
#5	Nephropathy OR Glomerulopathy OR Albuminuria OR Albumin-creatinine ratio OR Albumin creatinine ratio OR Albumin-to-creatinine ratio OR Albumin to creatinine ratio OR Albumin-excretion rate OR Albumin excretion rate OR Proteinuria OR Protein-creatinine ratio OR Protein creatinine ratio OR Protein-to-creatinine ratio OR Protein to creatinine ratio OR Protein-excretion rate OR Protein excretion rate OR Estimated glomerular filtration rate OR Glomerular filtration rate OR Kidney

	transplant OR Renal transplant OR Kidney transplantation OR Renal transplantation OR Kidney replacement OR Renal replacement
#6	#1 AND ((#2 AND #3 AND #4) OR #5)

Initial search date: 15 June 2021

Data Extraction

We extracted key data including first author, study name, country, study setting, study design, sample size, percentage male, average age, follow-up duration, definitions of CKD and glaucoma, effect size and covariates adjusted for.

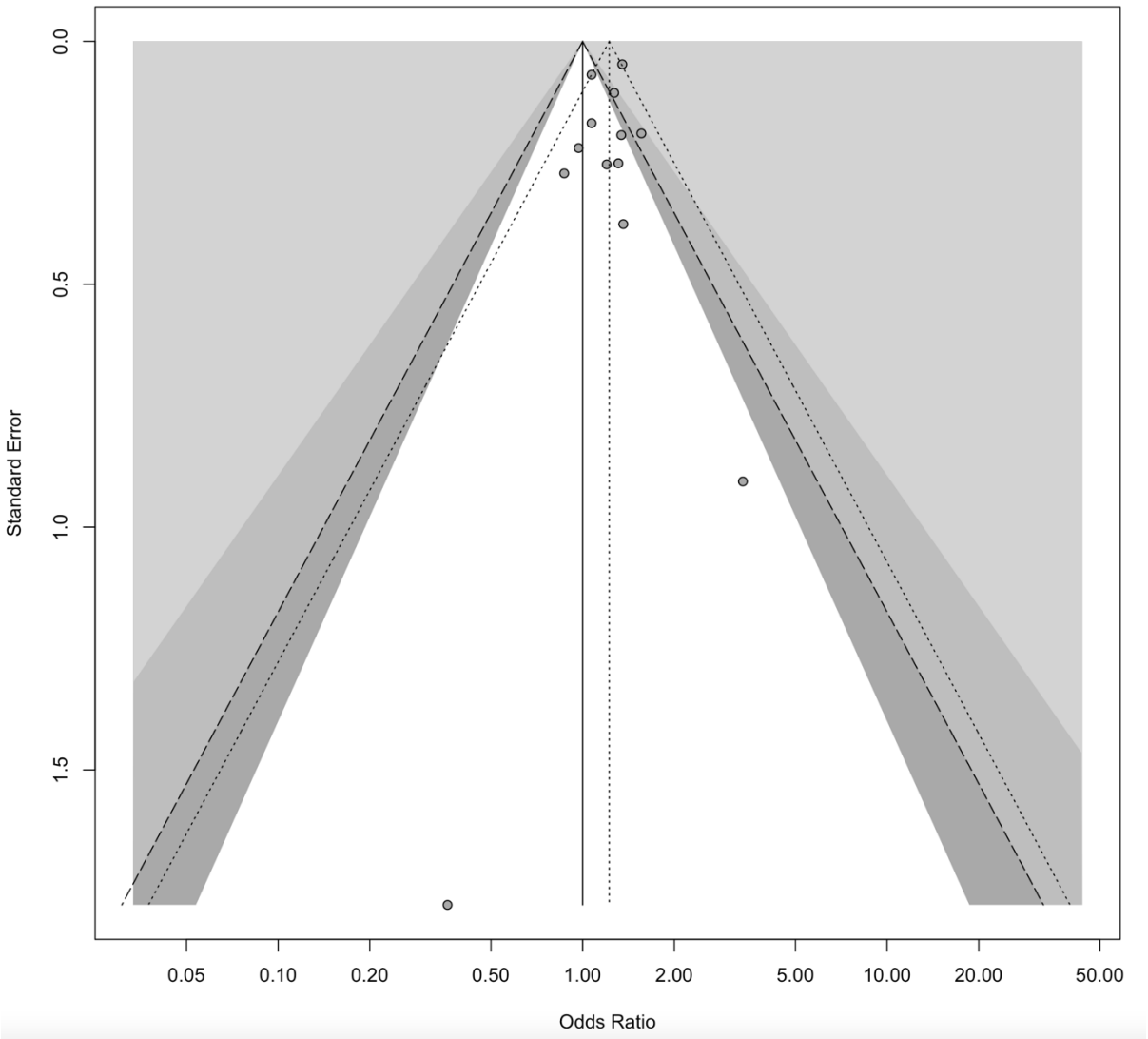
Statistical analyses

We used the mixed-effects model, consisting of a fixed-effects and random-effects model. If studies stratified their study population (e.g. by CKD severity), we pooled subgroups using a fixed-effects model to obtain an overall study summary estimate. We then pooled effect sizes at the study-level using a random-effects model to obtain an overall summary estimate. We pre-specified various study-level characteristics as potential explanatory variables including (1) mean age), (2) percentage male, (3) duration of follow-up, (4) number of covariates adjusted for, (5) setting/country, (6) study design, (7) risk of bias, (8) prevalence of comorbidities for further subgroup analyses and univariate random-effects meta-regression. We conducted all analyses using Review Manager (version 5.4.1) as well as R (version 4.0.3) using the *meta* package (version 4.18)

SUPPLEMENTAL FIGURES

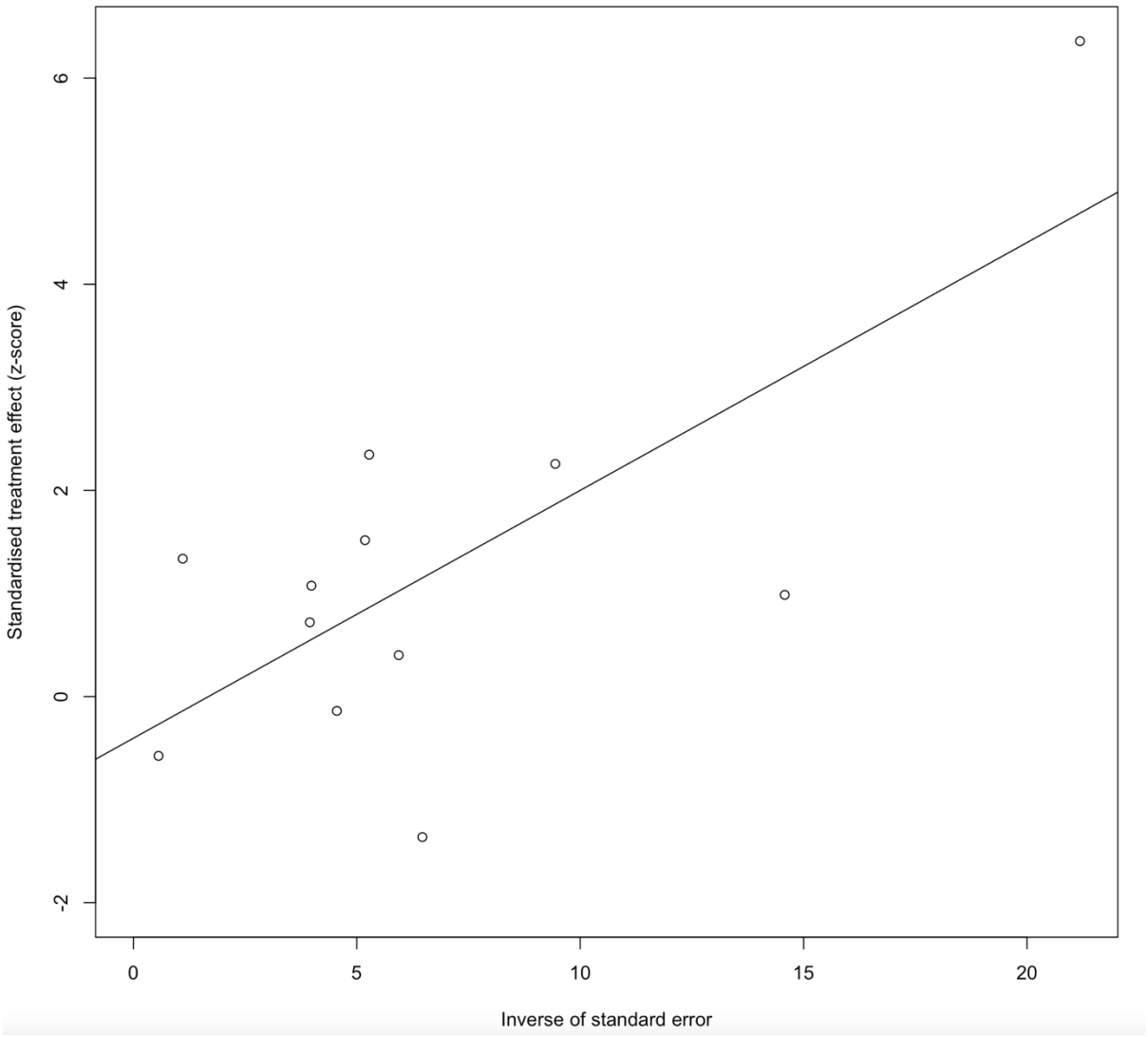
Supplemental Figure S1: Contour-enhanced funnel plot for studies reporting on the odds of glaucoma in CKD patients as compared to non-CKD patients, with missing studies imputed via the trim-and-fill method.

Legend: Shaded circles represent the original study estimates and the unshaded circle represents the missing estimate imputed via the trim-and-fill method. Dark gray, gray and light gray contour lines indicate conventional milestones in levels of statistical significance ($p<0.1$, 0.05 and 0.01).



*Upon visual inspection, there is no obvious funnel plot asymmetry in the distribution of studies. No additional studies were imputed via the Trim-and-Fill method

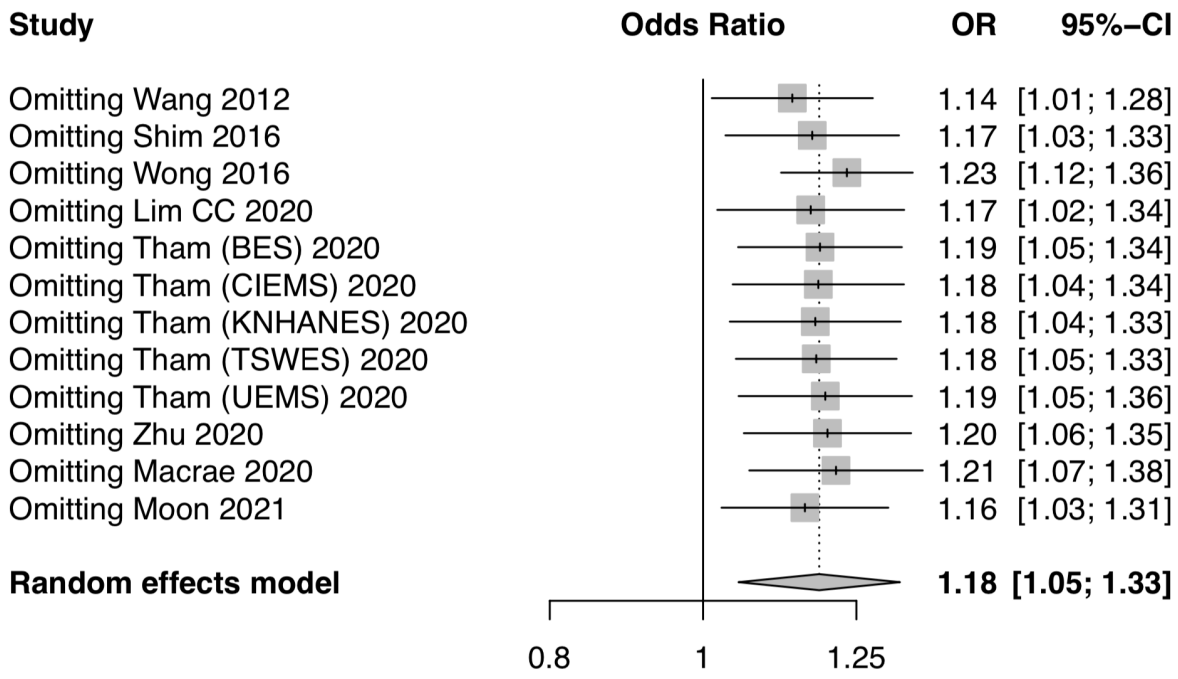
Supplemental Figure S2: Egger's bias for studies reporting on the odds of glaucoma in CKD patients as compared to non-CKD patients.



*The intercept of Egger's regression is 0.4598 (p = 0.5232), indicating that significant funnel plot asymmetry is unlikely

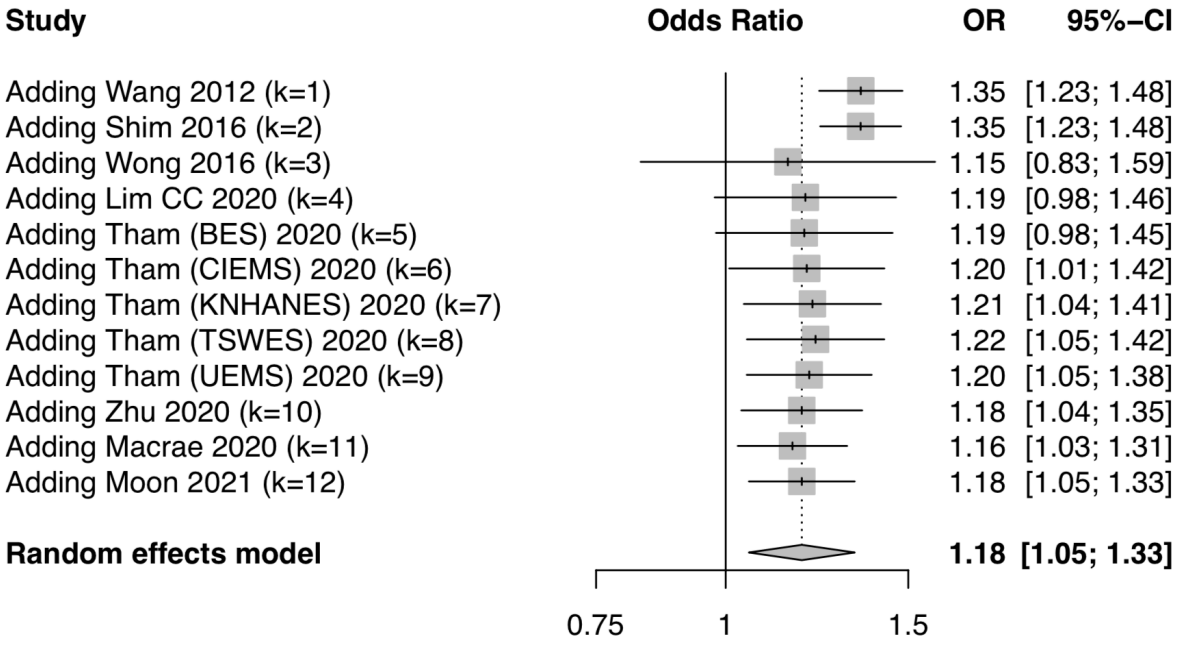
Supplemental Figure S3: Leave-out-one influence analysis for the odds of glaucoma in CKD patients as compared to non-CKD patients.

Legend: Gray diamonds are the estimated pooled hazard ratio (OR) for each random-effects meta-analysis; gray box sizes reflect the relative weight apportioned to studies in the meta-analysis.



Supplemental Figure S4: Cumulative influence analysis for the odds of glaucoma in CKD patients as compared to non-CKD patients.

Legend: Gray diamonds are the estimated pooled hazard ratio (OR) for each random-effects meta-analysis; gray box sizes reflect the relative weight apportioned to studies in the meta-analysis.



Supplemental Table S1: Risk-of-bias assessment of cohort and cross-sectional studies using the Newcastle-Ottawa Scale

Study	Representativeness of exposed group	Non-exposed group drawn from same community as exposed group	Ascertainment of exposure	Demonstrates that outcome of interest was not initially present	Adjusts for age	Adjusts for any comorbidity	Assessment of outcome	Median follow-up at least 5 years	Adequacy of follow-up	Total/9	Risk of bias*
Chou, 2018	1	1	1	1	1	1	1	1	0	8	Low
Lim CC, 2020	1	1	1	1	1	1	1	1	0	8	Low
Lim ZW, 2020	0	1	1	0	1	1	1	0	1	6	Moderate
Moon, 2021	1	1	1	1	1	1	1	1	0	8	Low
MacRae, 2021	1	1	1	1	1	0	1	0	1	7	Moderate
Nongpiur, 2010	0	1	1	0	1	1	1	0	1	6	Moderate
Park, 2019	1	1	1	1	1	1	1	1	1	9	Low
Shim, 2016	1	1	1	0	1	1	1	0	1	7	Moderate
Tham, 2020	1	1	1	0	1	1	1	0	1	7	Moderate
Wang, 2012	1	1	1	0	1	1	1	0	1	7	Moderate
Wong, 2016	1	1	1	0	1	1	1	0	1	7	Moderate
Yuksel, 2016	0	1	1	1	0	0	1	0	1	5	Moderate
Zakrzewski, 2012	0	1	1	0	1	1	1	0	1	6	Moderate
Zhu, 2020	1	1	1	0	1	1	1	0	1	7	Moderate

*8-9 stars: Low, 5-7 stars: Moderate, 0-4 stars: High

Supplemental Table S2: Evaluation of quality of pooled evidence using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) framework

Outcomes	Pooled outcomes (95% CI)	No. of participants (no. of included studies)	Statistical heterogeneity	Risk of bias	Imprecision	Inconsistency	Indirectness	Publication bias	Quality of evidence (GRADE)
Glaucoma	1.18 (1.04-1.33)	1,463,237 (12 studies)	I^2=68% (p<0.01)	Not serious	Not serious	Not serious	Not serious	Not serious	Low
CKD	2.44 (1.79-3.33)	508,673 (2 studies)	I^2=74% (p<0.01)	Not serious	Serious[2]	Serious[1]	Not serious	Serious	Very low

Quality of evidence for observational studies is graded starting at low quality for a causal effect, and downgraded or upgraded based on the following criteria: risk of bias, imprecision, inconsistency, indirectness and publication bias. 1. Downgraded by one level due to significant heterogeneity detected. 2. Downgraded by one level due to small number of studies pooled.

Supplementary Table S3: Random-effects meta-regression of log(ORs) against potential effect moderators (continuous and categorical study-level characteristics) for the odds of glaucoma in CKD participants

	Beta‡	**SE**	**Z**	**P**	**95% CI Lower**	**95% CI Upper**	**R^2 (% heterogeneity accounted for)**	**I^2 (% residual heterogeneity)**
Publication year	-0.0115	0.0167	-0.6927	0.4885	-0.0422	0.0211	32.13	38.39
Continent (Asia vs Non-Asia)	-0.1909	0.0940	-2.0306	0.1045	-0.3752	-0.0066	69.72	22.24
Ethnicity (East Asian vs Non-East Asian)	-0.2481	0.0494	-5.0211	**0.0061**	-0.3450	-0.1513	100.00	0.00
Study design (Cross-sectional vs Cohort)	0.1916	0.1608	1.1918	0.2333	-0.1235	0.5067	0.00	66.15
Sample size	-0.0000	0.0000	-0.9996	0.3175	-0.0000	0.0000	33.59	37.05
% male	0.0178	0.0110	1.6232	0.1045	-0.0037	0.0393	60.21	34.29
Mean age	-0.0219	0.0152	-1.4397	0.1500	-0.0516	0.0079	0.00	21.99
Number of covariates	-0.0069	0.0182	-0.3822	0.7023	-0.0425	0.0286	0.00	62.45
NOS Score	0.1916	0.1608	1.1918	0.2333	-0.1235	0.5067	0.00	66.15
Diabetes prevalence	0.0053	0.0033	1.5866	0.1126	-0.0012	0.0118	60.69	25.75
Hypertension prevalence	0.0006	0.0030	0.2012	0.8405	-0.0053	0.0065	9.38	46.65
Hyperlipidemia prevalence	-0.0078	0.0043	-1.8047	0.0711	-0.0164	0.0007	82.87	5.05
Cardiovascular diseases prevalence	-0.0006	0.0068	-0.0955	0.9239	-0.0140	0.0127	0.00	20.16

‡Estimated factor by which the log(OR) changes per unit increase in a continuous variable or in comparison with the reference group for a categorical variable. 95% Cis are also presented in the log scale